What is WiFi and How Does it Work?

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Wireless networks make it easy to connect to the internet wherever you go. BUSAKORN PONGPARNIT/GETTY IMAGES

There was a time when you'd only find WiFi at certain airports, coffee shops, or hotels. Nowadays, wireless internet connection, also known as WiFi or 802.11 networking, is in just about every home and business. Many wifi.htm">cities are also using the technology to provide free or low-cost internet access to residents.

WiFi has a lot of advantages. Wireless networks are easy to set up and inexpensive. They're also unobtrusive — unless you're on the lookout for a place to watch streaming movies on your tablet, you may not even notice when you're in a hot spot. In this article,

we'll look at the technology that allows information to travel over the air. We'll also review what it takes to create a wireless network in your home.

First, let's go over a few WiFi basics.

What Is WiFi?

A wireless network uses radio waves, just like cell phones, televisions and radios do. In fact, communication across a wi network is a lot like two-way radio communication. Here's what happens:

1. A computer's wireless adapter translates data into a radio signal and transmits it using an antenna.
2. A wireless router receives the signal and decodes it. The router sends the information to the internet using a physical, wired ethernet connection.

The process also works in reverse, with the router receiving information from the internet, translating it into a radio signal and sending it to the computer's wireless adapter.

The radios used for WiFi communication are very similar to the radios used for walkie-talkies, cell phones and other devices. They can transmit and receive radio waves, and they can convert 1s and 0s into radio waves and convert the radio waves back into 1s and 0s. But WiFi radios have a few notable differences from other radios:

- They transmit at frequencies of 2.4 GHz or 5 GHz. This frequency is considerably higher than the frequencies used for cell phones, walkie-talkies and televisions. The higher frequency allows the signal to carry more data.
- 2.4 GHz connections are now considered somewhat obsolete because they carry lower data speeds than 5 GHz. The 2.4 band continues to see use, however, because the lower frequency can carry over several hundred feet. In ideal conditions, the 5 GHz band has a max range of about 200 feet (61 meters), but in the real world, it is much more prone to interference from walls, doors and other objects. The 2.4

band may be faster for a user connecting to a router several rooms away, while 5 GHz will definitely be faster for a close connection.

WiFi uses 802.11 networking standards, which come in several flavors and have evolved over the decades:

802.11b (introduced in 1999) is the slowest and least expensive standard. For a while, its cost made it popular, but now it's less common as faster standards become less expensive. 802.11b transmits in the 2.4 GHz frequency band of the radio spectrum. It can handle up to 11 megabits of data per second, and it uses complementary code keying (CCK) modulation to improve speeds.

802.11a (introduced after 802.11b) transmits at 5 GHz and can move up to 54 megabits of data per second. It uses orthogonal frequency-division multiplexing (OFDM), a more efficient coding technique that splits that radio signal into several sub-signals before they reach a receiver. This greatly reduces interference.

802.11g transmits at 2.4 GHz like 802.11b, but it's a lot faster — it can handle up to 54 megabits of data per second. 802.11g is faster because it uses the same OFDM coding as 802.11a.

802.11n (introduced in 2009) is backward compatible with a, b and g. It significantly improved speed and range over its predecessors. For instance, although 802.11g theoretically moves 54 megabits of data per second, it only achieves real-world speeds of about 24 megabits of data per second because of network congestion. 802.11n, however, reportedly can achieve speeds as high as 140 megabits per second. 802.11n can transmit up to four streams of data, each at a maximum of 150 megabits per second, but most routers only allow for two or three streams.

802.11ac came on the scene around 2014, and operates exclusively at a 5 GHz frequency. 802.11ac is backward compatible with 802.11n (and therefore the others, too), with n on the 2.4 GHz band and ac on the 5 GHz band. It is less prone to interference

and far faster than its predecessors, pushing a maximum of 450 megabits per second on a single stream, although real-world speeds may be lower. Like 802.11n, it allows for transmission on multiple spatial streams — up to eight, optionally. It is sometimes called 5G because of its frequency band, sometimes Gigabit WiFi because of its potential to exceed a gigabit per second on multiple streams and sometimes Very High Throughput (VHT) for the same reason.

802.11ax, also known as WiFi 6, came to the industry in 2019. This standard extends the capabilities of 802.11ac in a few key ways. First of all, the new routers allow an even higher data flow rate, up to 9.2 Gbps (gigabits per second). WiFi 6 also lets manufacturers install many more antennas on one router, accepting multiple connections at once without any worry of interference and slowdown. Some new devices also connect on a higher 6 GHz band, which is about 20 percent faster than 5GHz in ideal conditions.

802.11be (or WiFi 7) started rolling out in 2024, offering even better range, more connections and faster data rates than any of the previous versions.

Other 802.11 standards focus on specific applications of wireless networks, like wide area networks (WANs) inside vehicles or technology that lets you move from one wireless network to another seamlessly.

WiFi radios can transmit on any frequency band. Or they can "frequency hop" rapidly between the different bands. Frequency hopping helps reduce interference and lets multiple devices use the same wireless connection simultaneously.

As long as they all have wireless adapters, several devices can use one router to connect to the internet. This connection is convenient, virtually invisible and fairly reliable; however, if the router fails or if too many people try to use high-bandwidth applications at the same time, users can experience interference or lose their connections, although newer, faster standards like 802.11ax will help with that.

Next, we'll look at how to connect to the internet from a WiFi hot spot.

Other Names and Other Wireless Network Standards

It's a common misconception that "Wi-Fi" is shorthand for "wireless fidelity", but the term actually comes from 802.11 networking standards. The 802.11 designation comes from the Institute of Electrical and Electronics Engineers (IEEE). The IEEE sets standards for a range of technological protocols, and it uses a numbering system to classify these standards.

WiFi Hot Spots



WiFi has enabled people to work anywhere they can get a signal, like at this coffee shop. MORSA IMAGES/GETTY IMAGES

A WiFi hot spot is simply an area with an accessible wireless network. The term is most often used to refer to wireless networks in public areas like airports and coffee shops. Some are free and some require fees for use, but in either case they can be handy when you are on the go. You can even create your own mobile hot spot using a cell phone or an external device that can connect to a cellular network. And you can always set up a WiFi network at home via Internet service providers.

If you want to take advantage of public Wi Fi networks or your own home-based network, the first thing you'll need to do is make sure your computer has the right gear. Most new laptops and many new desktop computers come with built-in wireless transmitters, and just about all mobile devices are WiFi enabled. If your computer isn't already equipped, you can buy a wireless adapter that plugs into the PC card slot or USB port. Desktop computers can use USB adapters, or you can buy an adapter that plugs into the PCI slot inside the computer's case. Many of these adapters can use more than one 802.11 standard.

Once you've installed a wireless adapter and the drivers that allow it to operate, your computer should be able to automatically discover existing networks. This means that when you turn your computer on in a WiFi hot spot, the computer will inform you that the network exists and ask whether you want to connect to it. If you have an older computer, you may need to use a software program to detect and connect to a wireless network.

Being able to get a wi-fi connection in public hot spots is extremely convenient. Wireless home networks are convenient as well. They allow you to easily connect multiple computers and other wireless devices and to move them from place to place without disconnecting and reconnecting wires. In the next section, we'll look at how to create a wireless network in your home.

Building a Wireless Network



Smartphones, tablets, smart TVs and laptops all access the internet with the help of a router (the device pictured with two antennas). PICTAFOLIO/GETTY IMAGES

If you already have several computers networked in your home, you can create a wireless network with a wireless access point. If you have several computers that are not networked, or if you want to replace your ethernet network, you'll need a wireless router. This is a single unit that contains:

- a port to connect to your cable or DSL modem
- a router, also known as a wireless local area network (WLAN)
- an ethernet hub
- a firewall
- a wireless access point

A wireless router allows you to use wireless signals or ethernet cables to connect your computers and mobile devices to one another, to a printer and to the internet. Most routers provide coverage for about 100 feet (30.5 meters) in all directions, although walls

and doors can block the signal. If your home is very large, you can buy inexpensive range extenders or repeaters to increase your router's range.

As with wireless adapters, many routers can use more than one 802.11 standard. Normally, 802.11n routers are slightly less expensive than others, but because the standard is older, they're also slower than 802.11ac or 802.11ax.

Once you plug in your router, it should start working at its default settings. Most routers let you use a web interface to change your settings. You can select:

- **The name of the network, known as its service set identifier (SSID).** The default setting is usually the manufacturer's name.
- **The channel that the router uses.** Most routers use channel 6 by default. If you live in an apartment and your neighbors are also using channel 6, you may experience interference. Switching to a different channel should eliminate the problem.
- **Your router's security options.** Many routers use a standard, publicly available sign-on, so it's a good idea to set your own username and password.

Security is an important part of a home wireless network, as well as public WiFi hot spots. If you set your router to create an open hot spot, anyone who has a wireless card will be able to use your signal. Most people would rather keep strangers out of their network, though. Doing so requires you to take a few security precautions.

It's also important to make sure your security precautions are current. The Wired Equivalency Privacy (WEP) security measure was once the standard for WAN security.

The idea behind WEP was to create a wireless security platform that would make any wireless network as secure as a traditional wired network. But hackers discovered vulnerabilities in the WEP approach, and today it's easy to find applications and programs that can compromise a WAN running WEP security. It was succeeded by the

first version of WiFi Protected Access (WPA), which uses Temporal Key Integrity Protocol (TKIP) encryption and is a step up from WEP but is also no longer considered secure.

To keep your network private, you can use one or more of the following methods:

- **WiFi Protected Access version 2** (WPA2) is the successor to WEP and WPA, and is now the recommended security standard for WiFi networks. It uses either TKIP or Advanced Encryption Standard (AES) encryption, depending upon what you choose at setup. AES is considered the most secure. As with WEP and the initial WPA, WPA2 security involves signing on with a password. Public hot spots are either open or use any of the available security protocols, including WEP, so use caution when connecting away from home. WiFi Protected Setup (WPS), a feature that ties a hard-coded PIN to the router and makes setup easier, apparently creates a vulnerability that can be exploited by hackers, so you may want to turn off WPS if possible, or look into routers that do not have the feature.
- **WPA3** was released in 2018 and became the security standard as of 2020. It aims to solve some of the vulnerabilities in WPA2 by putting much more complex encryption on both the router side and client side of the connection. This encryption also changes over time, meaning if a hacker had managed to access an unauthorized connection at one point in time, they would be locked out again the next time they try to connect. WPA3-enabled devices can also add some client-side encryption while using open public networks.
- It's worth noting that no matter how secure, a wireless network will surely have some method of exploit that can be used by hackers. When it comes to sensitive government or corporate data, a simple wired connection is the more secure alternative. In order to access or spy on a wireless network, a hacker has to be within physical range of the router, so attacks at home are not very likely to occur.
- **Media Access Control** (MAC) addresses filtering is a little different from WEP, WPA or WPA2. It doesn't use a password to authenticate users — it uses a computer's physical hardware. Each computer has its own unique MAC address. MAC address filtering allows only machines with specific MAC addresses to access the network. You must specify which addresses are allowed when you set up your router. If you buy a new computer or if visitors to your home want to use your network, you'll need to add the new machines' MAC addresses to the list of approved addresses. The system isn't foolproof. A clever hacker can spoof a MAC address —

that is, copy a known MAC address to fool the network that the computer he or she is using belongs on the network.

You also can change other router settings to improve security. For instance, you can set it to block WAN requests to keep the router from responding to IP requests from remote users, set a limit to the number of devices that can connect to your router and even disable remote administration so that only computers plugged directly into your router can change your network settings.

You should also change the Service Set Identifier (SSID), which is your network name, to something other than the default so that hackers can't immediately tell what router you are using. And selecting a strong password never hurts.

Wireless networks are easy and inexpensive to set up, and most routers' web interfaces are virtually self-explanatory. For more information on setting up and using a wireless network, check out the links that follow.

WiFi FAQ

What does a WiFi router do?

A wireless router provides internet access to every device thatâ€™s connected to it by creating a WiFi network using its built-in antennas. The router directly connects to a modem via a cable. The modem brings the internet service into your home. The router transmits the internet signal from the modem to all the devices connected in your home network.

How do I test my WiFi speed?

To test your WiFi speed, use the Google WiFi app (or another speed testing app or website) on whatever device you want to test the connect speed of. Each device that is connected to your WiFi has a different speed, which will be displayed in the app.

What is the difference between Google Nest WiFi and Google WiFi?

The Google Nest WiFi is faster, can handle more devices, has better coverage and looks sleeker (with a new dome shape) than the Google WiFi.

Can my WiFi be hacked?

Some hackers are efficient enough to hack into a WiFi connection. Change the Service Set Identifier (SSID), which is your network name, to something other than the default and set the router to block WAN requests to keep it from responding to IP requests from remote users.

Lots More Information

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- How Municipal WiFi Works
- How Home Networking Works
- What's the Difference Between a Modem and a Router?
- How Power-line Networking Works
- How Ethernet Works
- How Routers Work

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